

Digital Design & Verification Diploma

Why this diploma?

1. Students previously enrolled in the diploma and graduated were technically more ready for the market as per the industry's feedback [here](#)
2. The instructor has twelve years of teaching experience in addition to nine years of industry experience that will be shared with the students.
3. The diploma adds the following to your CV
 - Solid Knowledge of Siemens EDA simulation tool: QuestaSim
 1. Basic simulation usage
 2. Code Coverage
 3. Functional Coverage
 4. SystemVerilog Assertions
 - Solid Knowledge in Digital/RTL Design Basics
 - Solid Knowledge of Hardware Description Languages: Verilog & SystemVerilog
 - Solid Knowledge of UVM based verification methodology
 - Knowledge of one of the most used communication protocols (Diploma Project)
 - Solid Knowledge of Testing Strategies, Test Planning and Testing Automation
 - Solid Knowledge in assertion-based verification and functional coverage
 - Knowledge in Clock Domain Crossing techniques & Low power design implementations
 - Knowledge in Formal Verification - Static Linting, Formal Property Checking & Equivalence Checking
 - Knowledge of Siemens EDA formal tools: Questa Lint, Questa CDC & Questa RDC
 - Knowledge of FPGA design flow and architecture
 - Knowledge of Vivado design flow
4. Learn what really matters in terms of basics and language constructs required to get ready for the digital domain market.
5. Learn about the different job opportunities available in the digital industry market and the role of each title to give the students a more robust idea of what to expect after graduation.
6. Learn about the latest trends around the world in the design and verification industry.

Who is this diploma for?

1. Students in Communications/Electronics Engineering
2. Students in Computer/Software Engineering

Instructor

1. Basic Information & Education
 - Name: Kareem Waseem
 - Computer and System Engineering Master of Science - Ain Shams Uni
 - Electronics and Communications Engineering Bachelor - Ain Shams Uni
 - Ranked 1st
 - LinkedIn account: [here](#), Facebook group: [here](#)
2. Work Experience
 - **Co-Founder & Director of Engineer** at **SylvarisAI**, March 2025 - Present
 - **Digital Verification Senior Team Lead** at **Si-Vision**, September 2025 - March 2025
 - **Digital Verification Team Lead** at **Si-Vision**, March 2024 - September 2025
 - **Staff Verification Engineer** (Acting as team leader) at **Si-Vision**, March 2023 – March 2024
 - **Senior Hardware Modeling Engineer** at **Siemens EDA**, June 2022 – March 2023
 - **Hardware Modeling Engineer** at **Siemens EDA**, December 2017 – June 2022
3. Academic Mentorship for Students
 - **Si-Vision Digital Verification Graduation Project Mentorship** – 2025 & 2026
 - **Si-Vision Digital Verification Internship Mentorship** – 2024 & 2025
 - Internship focusing on advanced and trending topics in the industry (More detailed [here](#))
 - **Si-Vision Academy instructor** – 2023 - 2025
 - SystemVerilog language
 - SystemVerilog Assertions
 - **Siemens EDA Diploma instructor** – 2023
 - Verilog HDL
 - FPGA Design Concepts
 - Digital Verification using UVM

- **Teacher Assistant** for the **Electronic Design Automation (EDA)** course (Ain Shams University) - Fall 2023
- **Teacher Assistant** for the **Digital Verification** elective course (Ain Shams University) - Spring 2023
- **Teacher Assistant** for **ASIC Design & Automation** elective course (Ain Shams University) - Fall 2022 & Fall 2023
- **Lecturer and Teacher Assistant** for the **Digital Verification** elective course (Ain Shams University) - Spring 2021

DIGITAL DESIGN USING VERILOG AND FPGA FLOW USING VIVADO

Topics Covered

1. Classical Digital Design flow – Truth tables/K-maps/Boolean Algebra/Logic Equivalence
2. History of HDLs & Intro to Verilog/Verilog Constructs [Modern Digital Design flow]
3. Combinational Logic basics & Verilog Essentials to model Combinational Logic circuits
4. Key combinational logic circuits and their implementation in Verilog
5. Verification using Verilog, including testbench development
6. Sequential circuits basics & Verilog essentials to model sequential circuits
7. Key sequential circuits and their implementation in Verilog
8. Finite State Machine (FSM) basics/types & how to design them in Verilog
9. Memories basics/types & Verilog essentials to model memories
10. Key memory types and their implementation in Verilog
11. Verilog synthesis constructs
12. Introduction to widely used communication protocols and interfaces, covering foundational concepts and highlighting commonalities across standards.
13. Tcl commands to automate Siemens QuestaSim simulation flow
14. Synthesis & timing fundamentals, including Static Timing Analysis (STA), and pipelining concepts.
15. Static Linting Checks & Clock Domain Crossing (CDC) techniques
16. Applying Static Checks using Siemens QuestaFormal tools
17. Low power design implementations
18. FPGA design flow & architecture basics
19. Vivado design flow using Basys 3 FPGA Board

20. Timing and physical design constraints
21. Integrate IP cores into design flow using IP Catalog
22. FPGA-based Prototyping & Partitioning Challenges
23. Course Projects (Two projects)

Advanced Topics / Post-Diploma Extension

1. More static timing analysis techniques
2. Hardware/Software co-design basics

Note: These advanced topics are planned as optional post-diploma sessions for students who would like to continue beyond the core diploma. The main diploma will continue to focus on building strong fundamentals first. These optional advanced topics are expected to take around 1 week and will be delivered by Eng. Kareem Waseem and other engineers currently working in the industry, under Eng. Kareem's supervision and attendance, to ensure consistent technical quality.

Pre-requisite

1. Digital Logic Basics

Fees & Course duration

1. Course fees for **students** or **unemployed**: 5000 LE.
2. Course fees for **employed graduates**: 6000 LE.
3. Course fees **include** center fees, hard copy notes and certificate "Softcopy and hardcopy"
4. Course duration: 6 weeks (36 hours)
 - Please note that the course duration may be extended in case of emergencies that result in a cancelled session, or if additional time is needed based on the pace of the class.
5. Course starts on June 26th, 2026
 - One class per week
 - Friday 10 pm – 1 pm then 1 hour break (Jumu'ah prayers) then (2 pm – 5 pm)
 - i. 4 hours of lecture
 - ii. 2 hours of mini guided hands-on lab work, divided into sections and led by my top-trained assistants under my supervision, with exercises based directly on the lecture material.

Reservation Rules

- Deadline for payment: June 19th, 2026

DIGITAL VERIFICATION USING SYSTEMVERILOG AND UVM

Topics Covered

1. Verification Trends, Planning & Latest Findings
2. SystemVerilog Data Types & Subroutines
3. Code Coverage
4. SystemVerilog Interfaces
5. SystemVerilog Assertions & Threads
6. SystemVerilog Scheduling Semantics
7. Basics of Object-Oriented Programming
8. Constrained Random Value Generation
9. Functional Coverage
10. UVM Structure overview
11. UVM Components and Phases
12. UVM Component Communication
13. UVM Sequence & Sequence item
14. UVM Configuration & Factory
15. Formal Verification using formal property checking & equivalence checking
16. Introduction to hardware-assisted verification platforms (FPGA-based prototyping & Emulators)
17. Introduction to the current latest trends for digital verification
18. Introduction to current AI applications for digital design & digital verification
19. Course Project (SystemVerilog Project and UVM Project)

Advanced Topics / Post-Diploma Extension

1. SystemVerilog DPI-C
2. Bus Functional Models
3. Reactive UVM agent techniques
4. Virtual sequences and virtual sequencers
5. Introduction to UVM RAL

6. Advanced UVM TLM communication
7. Regression testing techniques
8. Variable lifetime: static and automatic
9. Dynamic processes: (disable fork/ fork guard), and forks with loops
10. Inter-process synchronization and communication: mailbox and semaphore
11. Data hiding in OOP
12. Pure virtual methods and abstract classes
13. Clocking blocks and program blocks
14. Advanced constrained randomization techniques
15. More SystemVerilog assertion operators and dynamic assertion enablement
16. File handling

Note: These advanced topics are planned as **optional post-diploma sessions** for students who would like to continue beyond the core diploma. The main diploma will continue to focus on building strong fundamentals first. These optional advanced topics are expected to take around **4 weeks** and will be delivered by Eng. Kareem Waseem and other engineers currently working in the industry, under Eng. Kareem's supervision and attendance, to ensure consistent technical quality.

Pre-requisite

1. Verilog with basic understanding of designing and verifying a digital circuit using digital simulator.
2. Engineers with no Verilog knowledge are advised to enroll in the "Digital design using Verilog and FPGA flow using Vivado" mentioned in the beginning of this document.

Note: You are **not** required to have object-oriented programming background.

Fees & Course duration

1. Course fees for **students** or **unemployed**: 5500 LE.
2. Course fees for **employed graduates**: 6500 LE.
3. Course fees **include** center fees, hard copy notes and certificate "Softcopy and hardcopy"
4. Course duration: 7 weeks (42 hours)
 - Please note that the course duration may be extended in case of emergencies that result in a cancelled session, or if additional time is needed based on the pace of the class.
5. Course starts on August 7th, 2026

- One class per week
- Friday 10 pm – 1 pm then 1 hour break (Jumu'ah prayers) then (2 pm – 5 pm)
 - i. 4 hours of lecture
 - ii. 2 hours of mini guided hands-on lab work, divided into sections and led by my top-trained assistants under my supervision, with exercises based directly on the lecture material.

Reservation Rules

- Deadline for payment: July 31th, 2026

GENERAL & COMMON LOGISTICS

1. You must Fill in this [Google form](#) (ignore if you have filled it).
 - You will be contacted via the email you provided once submitting the form to complete the payment.
 - There are limited spots available for the diploma. Please check my facebook group [here](#) if there is an announcement that the diploma is complete before paying the fees.
2. Location
 - VIP center
 - عنوان السنتر: السراج مول - برج 1 - مدخل 1 - الدور الأول - بجوار ماكدونالدز - مدينة نصر
 - Google Maps link [here](#)
3. Each Class will have the following
 - Labs to encourage on-time application of the topic.
 - Assignment to be submitted.
 - Discuss real Interview Question/s from the industry to be better aware of the questions/topics that the industry focuses on (Mostly from Interviews done in Egyptian Companies)

DIPLOMA OFFER & GENERAL RULES

Diploma Offer

- Enroll in the two courses mentioned above as one diploma for 9000LE and save 1500LE, as opposed to paying 5000 and 5500 for the two courses separately.
- Diploma offer is for **students** or **unemployed**

- Deadline for diploma offer payment: June 19th, 2026

General Rules

1. Fees are 100% refundable until the fees deadline date
2. Fees are 90% refundable after the fees deadline and before the course starts
3. Fees are not refundable once the course starts
4. You cannot take just one course after paying for the diploma offer mentioned above.
5. In case you enroll in the diploma offer you will attend the two courses right after each other. You cannot postpone and attend the second course in later diplomas.
6. In case of missing a session, it is totally acceptable to attend it in the following diplomas.

Instructor: Kareem Waseem, LinkedIn [link](#), Facebook group [link](#)